Rohan **Dubey**

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EDUCATION

BIRLA INSTITUTE OF TECHNOLOGY SCIENCE, PILANI M. Sc. MATHEMATICS, B.E. IN ELECTRONICS AND ELECTRICAL ENG.
Expected Aug 2023 |

LINKS

Github:// **rohandubey**
LinkedIn:// **rohandubey**

COURSEWORK

UNDERGRADUATE

Discrete Mathematics
Optimization, Operations Research
Graphs and Networks
Non-Linear Optimization
Game Theory
Digital Design
Numerical Analysis
Signals and Control Systems
Ordinary Differential Equations

SKILLS

PROGRAMMING

Over 5000 lines

- Python
- C/C++

Over 1000 lines:

- SQL
- Matlab
- HTML/CSS
- Shell
- Julia

FRAMEWORKS

- Cloud : GCP(Deployment and Mitigation), GCP(Serverless Architecture)
- Kubernetes, Docker
- Flask, YOLO, CUDA, Libraries(OpenCV, PIL, Graphviz, Tensorflow, Pytorch)
- HTTP, TCP/IP, RTSP, RTMP

OPERATING SYSTEMS

Linux/Unix, MacOS, Windows, Git

EXPERIENCE

Andhra Pradesh Human Resource Development Institute | SUMMER INTERN
18 May 2020 – 27 June 2020

- Qualitative and Quantitative Research on COVID-19 in India.
- Predicted and Plotted the COVID-19 Data using Machine learning tools using SEIR model.

Indian Institute of Technology(BHU), Varanasi | SUMMER INTERN
20 May,2019 - 15 July, 2019

- Simulation of weighted Cost-Sharing Network Game in Multi-cast Routing Networks.
- Planned out network for a hypthetical city using Modified Nash Equilibrium and Graph and Network Theory.

PROJECTS

FACE MASK DETECTOR

- Based on YOLOv5 and RetinaNet Model, this custom computer vision model detects whether people are wearing a mask or not in a crowded place.
- This software is specifically designed to be installed in Police Control Rooms to monitor and collect data.
- The Software is tested by Rajasthan Police IT Centers and will be adopted by the Government and installed across state.
- Tech Stack: PyTorch, OpenCV, ONXX, Machine Learning and CUDA/CUDNN

PULSE-RATE MONITORING SYSTEM

- A software that detects the Pulse Rate of an individual using a webcam.
- This software is made for touchless detection and uses techniques like FFT and Filtering.
- Tech Stack: OpenCV, Signals Processing, Optimization

OBJECT DETECTION by HOMOGRAPHY

- A software that detects the given object in a live video feed with the help of KNN and FLANN algorithm.
- Tech Stack: Machine Learning, OpenCV

MINOR PROJECTS

- Numerical Analysis techniques
- OpenCV Mouse interface
- Feature Detection Software
- Object Optical Tracking
- Label Data Helper

CO-CURRICULAR

- Esports Club, BITS Hyderabad (Media Designer)
- IEEE student chapter (Technical Team Member)
- NSS, BITS Hyderabad (Student Volunteer)

SKILL-RELEVANT COURSEWORKS

- Cloud Engineering Track and Data Science & Machine Learning Track (Qwiklabs)
- Machine Learning Foundations (University of Washington | Coursera)
- Introduction to TensorFlow (DeepLearning.AI | Coursera)
- Computer Vision (Basics & Advanced) (Coursera)